

Rishikanth Manimeli

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Education

University of Massachusetts Amherst <i>MS in Computer Science GPA: 4.0/4.0</i>	01/2025 – 12/2026
Indian Institute of Technology Patna <i>Bachelor of Technology in Electrical Engineering</i>	07/2018 – 05/2022

Experience

Samsung R&D

<i>Senior Software Engineer C++, GTest, Python, Linux</i>	06/2022 – 01/2025
- Optimized MAC Uplink scheduling for Cell on Wheels deployments across Telus and Verizon modems, increasing cell capacity and LTE throughput by 25%. - Profiled and optimized core C++ system modules in a Linux environment to resolve CPU performance bottlenecks, reducing cycle consumption and improving throughput by 15%. - Built automated log analysis application in Python, reducing manual log analysis time from 30 minutes to 5 minutes. - Drove architectural improvements to the modem scheduler; contributed 100+ commits and 300+ code reviews over two years across the modem software stack.	

Projects

Agentic RAG Research Extern <i>Industry Experience at Adobe</i>	01/2026 – 05/2026
- Built a multi-document question-answering pipeline using FlashRAG, indexing 1,000+ research papers and internal documents with sub-second retrieval latency. - Developed a multi-hop reasoning benchmark on the Enron email corpus to robustly evaluate agentic retrieval; since LLMs are pretrained on publicly available data, existing benchmarks suffer from data pollution where models answer from memorized knowledge rather than retrieved documents. - Created an agentic RAG workflow using tool calling to retrieve missing context mid-inference, reducing hallucination rates on internal benchmarks.	

In-Memory Key-Value Store | *Go*

- Engineered a Redis-compatible in-memory data store in Go, implementing the RESP (REdis Serialization Protocol) parser and TCP networking to support concurrent client connections via goroutines.
- Built a leader-follower replication system supporting full state synchronization and real-time command propagation across replica nodes.
- Implemented robust storage features including key expiration (TTL) scheduling and RDB file parsing for persistent storage recovery.

Stock Market Application | *Python, AWS*

- Built a fault-tolerant microservices trading platform on AWS; implemented LRU caching with server-push invalidations and leader-based replication for automatic failover.
- Implemented Raft and Multi-Paxos consensus algorithms to guarantee totally ordered writes across replicas and tolerate node failures using a majority quorum.
- Load-tested the system under concurrent multi-user traffic, validating consistent data integrity, low latency, and correct cache behavior at scale.

Database Implementation | *Java*

- Engineered a custom relational database management system (DBMS) from scratch in Java, implementing core storage components including a buffer pool manager, on-disk page management, and a heap file engine.
- Implemented a B+ tree index structure, improving database lookup times from $\mathcal{O}(N)$ to $\mathcal{O}(\log N)$, and an iterator-based query executor supporting Block Nested Loop (BNL) joins.

Technologies

Programming Languages: C++, Python, Java, Go, Rust, C, TypeScript
Software Development & Tools: AWS, Docker, Git, Linux, PostgreSQL, MongoDB, REST APIs
Frameworks & Libraries: Qt, GTest, FlashRAG
Domains & Methodologies: Large Language Models, Agentic AI, Distributed Systems, Machine Learning, NLP

Leadership & Achievements

Course Grader & Project Mentor (COMPSCI 426): Managed Agile sprints for an 8-student team building an event-ticketing application, evaluating system architecture and functional correctness (Spring 2026).
Samsung Excellence Award: Recognized for exceptional contributions to system capacity improvement and core code optimization (April 2023).
Software Excellency Certification: Earned Samsung's Professional-level certification in algorithmic problem solving (October 2023).